

LAB–8: QUEUE APPLICATIONS

Objective

To implement various **applications of Queue and Deque** and analyze their efficiency in solving practical problems.

1:Reverse a Queue

Aim

To reverse the elements of a queue.

Input

Queue:

1 → 2 → 3 → 4

Output

4 → 3 → 2 → 1

Expected Approach (Method 1: Using Stack)

1. Dequeue all elements from queue and push them into a stack.
2. Pop elements from stack and enqueue back into queue.
3. The queue will now be reversed.

Expected Approach (Method 2: Using Recursion)

1. Dequeue the front element.
2. Recursively reverse the remaining queue.
3. Enqueue the removed element at the rear.

Time Complexity

$O(n)$

2: Count Number of Elements in Queue

Aim

To count the total number of elements present in a queue.

Expected Approach (Array Implementation)

If queue is implemented using array:

```
[  
Count = rear - front + 1  
]
```

Expected Approach

- Traverse queue without disturbing order
- Use a temporary queue if needed

3: Find Maximum and Minimum Element in Queue

Input

Queue: 4, 7, 1, 9

Output

Max = 9

Min = 1

Expected Approach

1. Traverse queue from front to rear.

2. Keep track of:
 - Maximum element
 - Minimum element
3. Ensure queue order is restored after traversal.

4: Check Palindrome Using Deque

Aim

To check whether a string is a palindrome using a double-ended queue (Deque).

Input

madam

Output

Palindrome